

Age- and Arm-Stratified Co-expression Network Rewiring in the Murine Kidney During and After Spaceflight

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Background & Motivation

Astronauts face a $\sim 5\times$ **elevated risk of nephrolithiasis** post-flight, with distal convoluted tubule (DCT) ion-transport dysregulation strongly implicated [1]. Standard differential expression (DE) detects only mean expression shifts—**regulatory rewiring** (changes in co-expression architecture) is not captured when mean levels are stable.

Central question: Does spaceflight restructure the kidney co-expression network in an age- and arm-dependent manner, producing transcriptomic adaptations that escape DE?

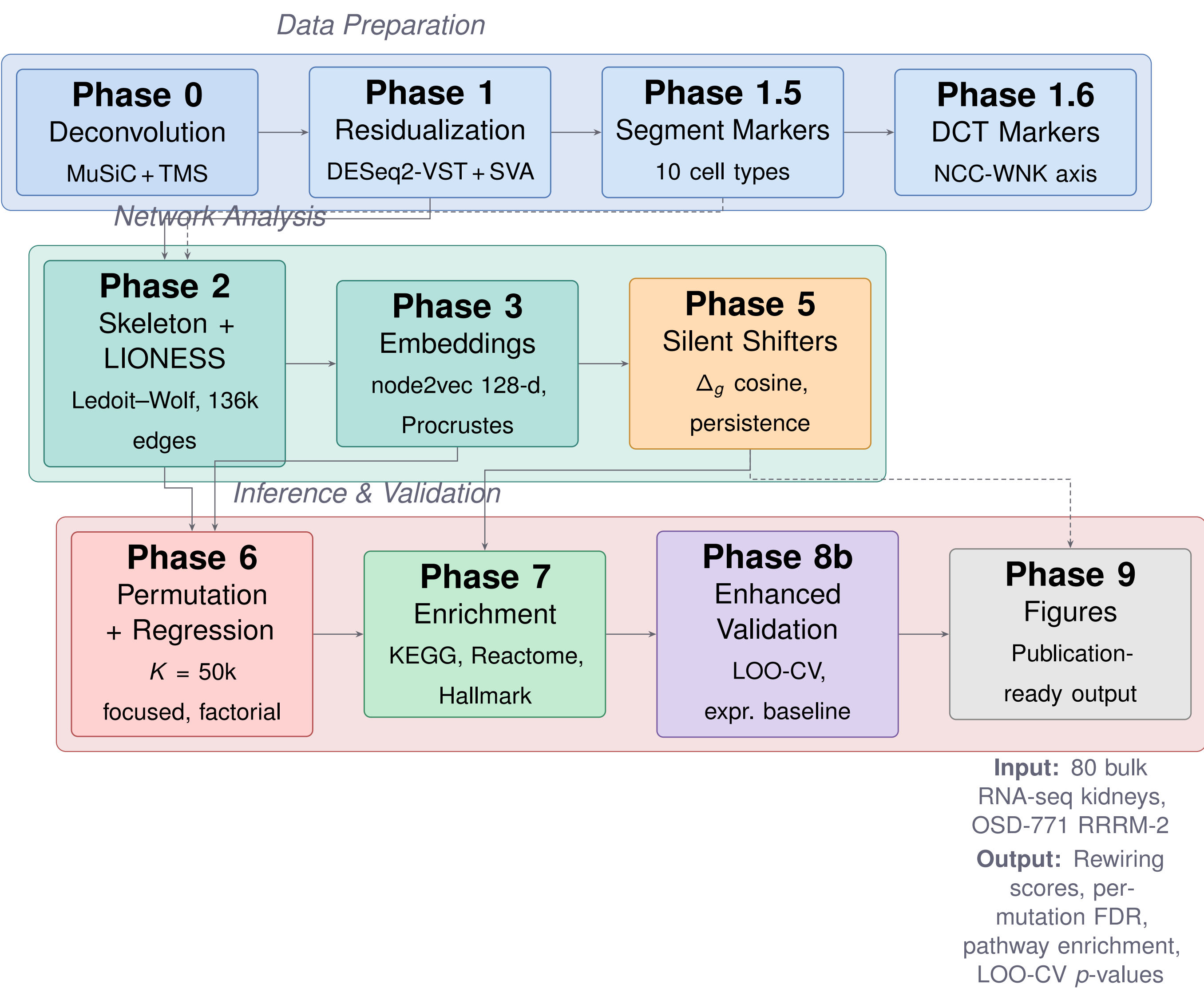
We developed a multi-phase pipeline integrating cell-type deconvolution (MuSiC), sample-specific network inference (LIONESS), graph embedding (node2vec with Procrustes alignment), and permutation testing applied to **RRRM-2 (OSD-771)**—the largest full-factorial spaceflight kidney RNA-seq dataset ($n = 80$).

Experimental Design: OSD-771 (RRRM-2)

Factor	Levels
Age	Young (16 wk) vs Old (34 wk)
Arm	ISS-T (terminal) vs LAR (live return)
Group	FLT GC VIV BSL
Per-cell n	5 mice $\times$ 2 ages = 10 per cell
Total n	80 female C57BL/6NTac bulk kidney RNA-seq

Four primary contrasts (FLT–GC): ISS-T Young, ISS-T Old, LAR Young, LAR Old. Arm stratification enables classification of rewiring as *ephemeral* (ISS-T-only) vs. *persistent* (both arms).

Multi-Phase Analytical Pipeline



Conclusions

- Spaceflight induces age-stratified co-expression rewiring orthogonal to DE ($\rho \approx 0.03$, $R^2 < 0.3\%$)
- 55–68 silent shifters per contrast**—connectivity restructuring with no detectable expression change
- Young ISS-T: ECM/complement rewiring; Old ISS-T: ribosomal/proteostasis disruption

Genome-Wide Rewiring Landscape

Spaceflight remodels co-expression architecture across all four contrasts. Mean rewiring $\Delta \approx 0.27$ – 0.29 (cosine distance); maximum single-gene rewiring exceeds 0.53.

Contrast (FLT–GC)	Mean Δ	Max Δ
ISS-T Young	0.284	0.515
ISS-T Old	0.287	0.511
LAR Young	0.271	0.513
LAR Old	0.275	0.539

Orthogonality: Rewiring Δ is independent of $|\log_2 FC|$ across all four contrasts ($\rho = 0.02$ – 0.05 , $R^2 < 0.3\%$), confirming that network rewiring captures a dimension of spaceflight biology orthogonal to differential expression.

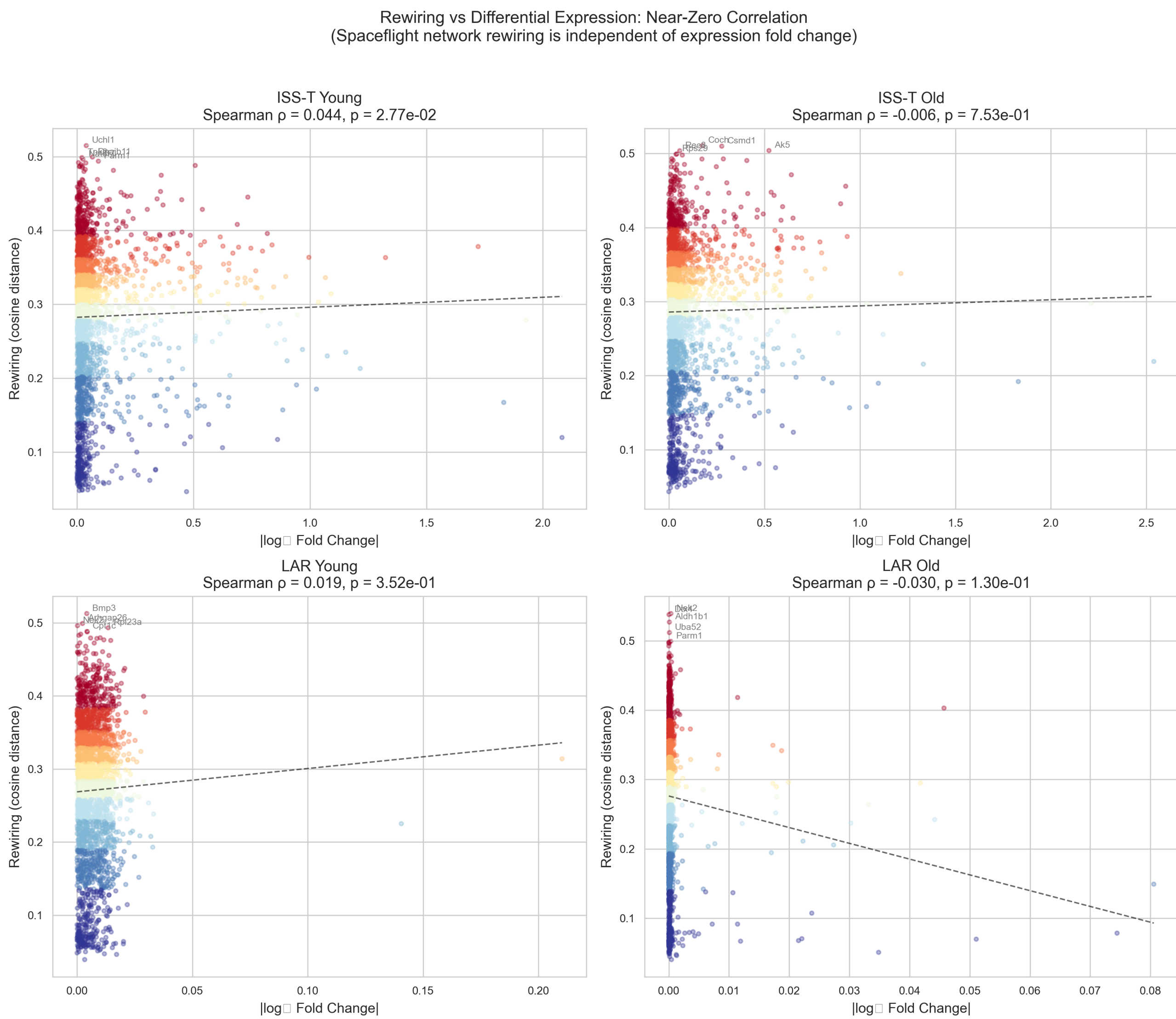
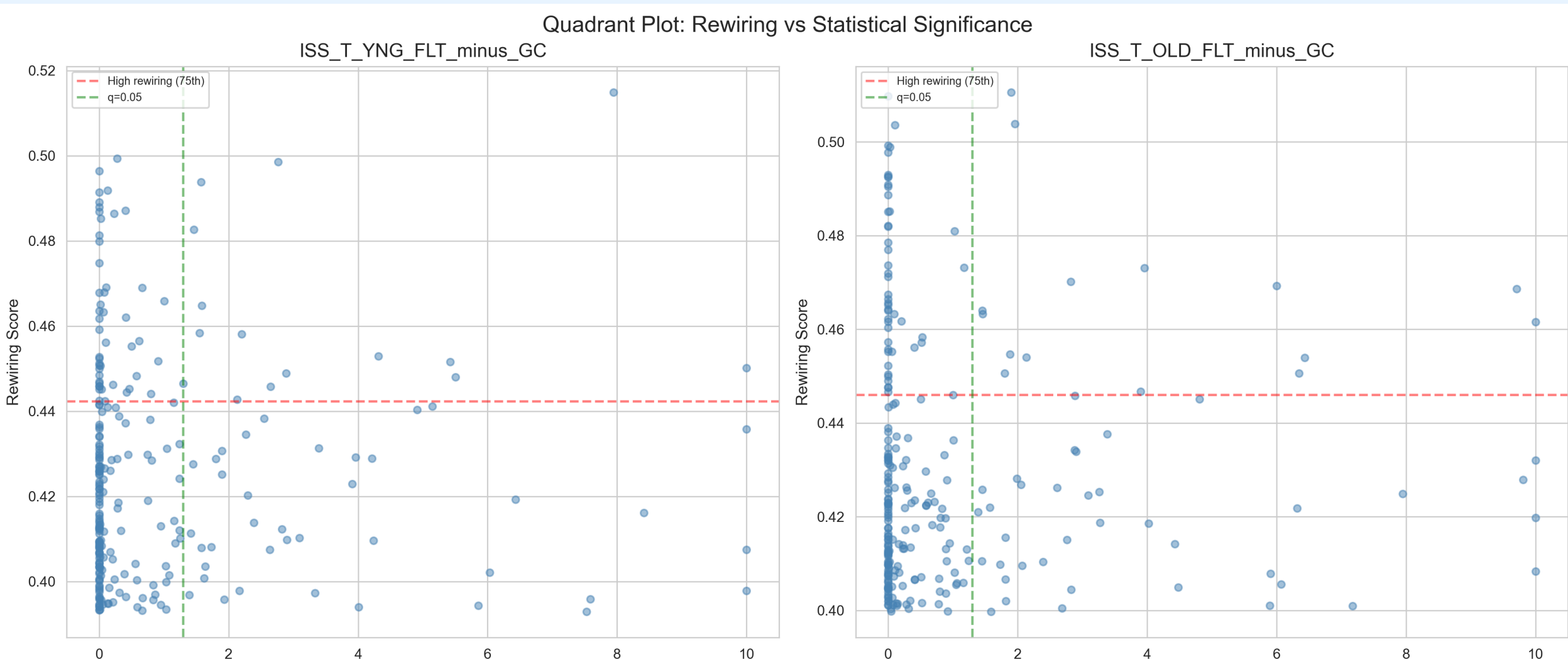


Figure 1. Rewiring magnitude (Δ) vs. $|\log_2 FC|$ for 2,500 genes across four contrasts. Near-zero Spearman ρ ($R^2 < 0.3\%$) confirms statistical independence of network rewiring from mean expression change.

Silent Shifters: Invisible Transcriptomic Restructuring

Silent shifters—genes in the top-decile of rewiring with $|\log_2 FC| < 0.3$ and DE FDR > 0.2 —are invisible to standard DE analysis. The pipeline identifies **55–68 silent shifters per contrast**:

Contrast	Silent Shifters
ISS-T Young	66
ISS-T Old	63
LAR Young	55
LAR Old	68



Age-Dependent Rewiring Divergence

ISS-T rewiring is correlated across age groups (Spearman $\rho = 0.855$), but a subset of genes shows age-divergent topological responses. The top age-divergent genes include Avpi1, Cxcl14, Parm1, Ak5, and Mcm3, spanning ECM remodeling, immune signaling, and cell-cycle regulation.

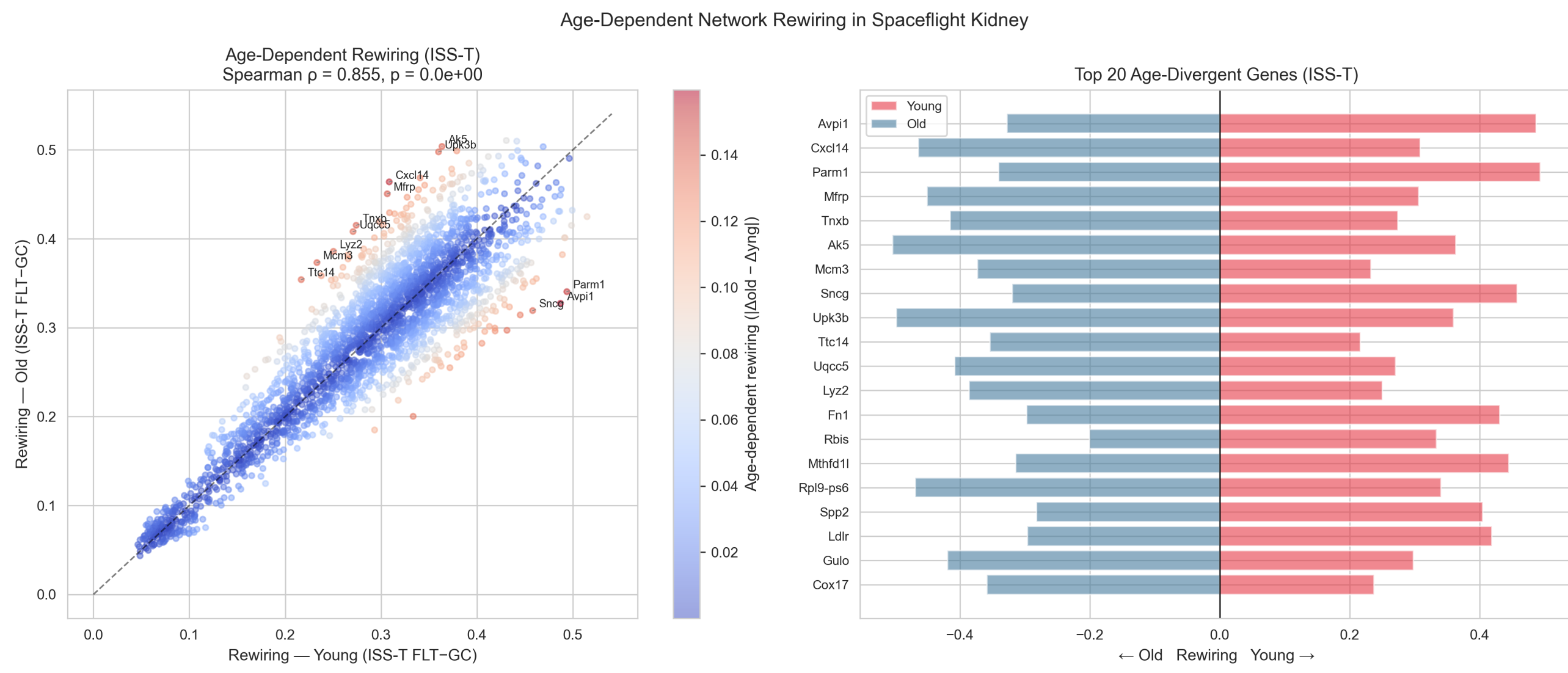


Figure 3. *Left:* ISS-T Old vs. Young rewiring scatter ($\rho = 0.855$). Color encodes age-divergent rewiring magnitude; labeled outliers are the most age-divergent genes. *Right:* Top 20 age-divergent genes, directional rewiring bar chart.

Pathway Remodeling: Age-Dependent Signatures

Gene set enrichment (Fisher's exact, BH-corrected; KEGG / Reactome / Hallmark / WikiPathways) reveals contrast-specific pathway signatures:

ISS-T Old (27 pathways, FDR < 0.05): Ribosome, Peptide Chain Elongation, Chaperone-Mediated Autophagy (OR = 23.6, $q = 0.005$), Selective Autophagy $\rightarrow$ *proteostasis disruption in the aged spaceflight kidney*

LAR Old (22 pathways): PPAR Signaling (OR = 7.5, $q = 0.027$), NOTCH Signaling, Complement/Coagulation $\rightarrow$ *coordinated metabolic and proteostatic recovery upon return*

LAR Young (11 pathways): PPAR Signaling (OR = 10.1, $q = 0.002$): Acaa1b, Acox2, Adipoq, Angptl4, Fabp5, Hmgcs2, Scd1 $\rightarrow$ *PPAR-centered metabolic recovery program*

